Suvinava Basak

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Education

M.Sc. **Technische Universität Braunschweig**, Data Science — Braunschweig, Germany, Oct 2023 – present
- Current average: 1.3 (German scale, 1.0 = best), excluding thesis (due Oct 15, 2026).
- Thesis: End-to-End Calibration: Theory and Empirical Evaluation in Biomedical Context.

MBA **Indian Institute of Management Indore**, Business Administration — Indore, India, July 2016 – Mar 2018

B.Tech. **Jalpaiguri Government Engineering College (West Bengal University of Technology)**, Computer Science & Engineering — West Bengal, India, Aug 2010 – June 2014
- DGPA: 7.83/10. Project: Face Recognition System using MATLAB.

Experience

TU Braunschweig, Institute for Statistics and Foundations of Machine Learning, M.Sc. thesis: End-to-End Calibration, Theory and Empirical Evaluation in Biomedical Context — Braunschweig, Germany, Apr 2026 – present, 7 months
- Investigating end-to-end calibration of neural networks, incorporating calibration objectives directly into model training rather than post-hoc calibration, and empirically evaluating its effect on predictive reliability and uncertainty estimation. Supervised by Prof. Dr. Nicole Mücke.
- Studying the relationship between predictive performance, confidence and reliability under distribution shift, using Transformer-based gene regulatory network inference as application.
- Designing and implementing Python/PyTorch experiments to evaluate neural-network calibration methods on biological regulatory network data.

TU Braunschweig, Institute for Statistics and Foundations of Machine Learning, Student Research Assistant (Student Hilfswissenschaftler) — Braunschweig, Germany, Mar 2026 – May 2026, 3 months
- Conducted research on calibration and uncertainty estimation in modern neural networks under the supervision of Prof. Dr. Nicole Mücke.

Medizinische Hochschule Hannover, Student Research Assistant (Student Hilfswissenschaftler) — Hannover, Germany, Aug 2025 – present, 1 year 3 months
- Developing an LLM- and Knowledge Graph-aware RAG based multi-agent system for scientific hypothesis generation from heterogeneous biomedical evidence; supervised by Dr. Zahra Ahmadi.
- Investigating evidence-based reasoning, calibrated confidence, and risk-controlled mechanisms for AI-assisted scientific reasoning under incomplete or heterogeneous evidence.

TU Braunschweig, Institut für Analysis und Algebra, Teaching Assistant (Student Hilfswissenschaftler) — Braunschweig, Germany, Apr 2025 – Sept 2025, 6 months
- Led tutorials and graded assignments for Mathematical Algorithms and Programming 2 (SS 2025).

Volkswagen Financial Services AG, Intern, Data Analytics & AI — Braunschweig, Germany
Aug 2024 – Feb 2025
7 months

- Built an ML pipeline automating invoice analysis and approval in vehicle maintenance workflows, reducing manual review time; designed a multi-stage Docker setup improving container security and pipeline build time.

Wipro Ltd., Strategy Consultant — Bangalore, India
June 2018 – Oct 2023
5 years 5 months

- Led AI/ML innovation scouting and startup partnerships, from technical evaluation through joint proof-of-concept development and go-to-market planning.

Ericsson Global India Services Pvt. Ltd., Solution Integrator — Kolkata, India
Mar 2015 – June 2016
1 year 4 months

- Configured and integrated a billing system for a leading US-based mobile carrier, improving end-to-end system consistency; analyzed historical usage patterns to design billing plans aligned with client offerings and user behavior.

Publications

From literature to hypotheses: An AI co-scientist system for biomarker-guided drug combination hypothesis generation
Raneen Younis, Suvinava Basak, Lukas Chavez, Zahra Ahmadi
arxiv.org/abs/2603.00612

Is grokking a loss of normal hyperbolicity of the interpolation manifold?
Suvinava Basak
arxiv.org/abs/2608.14803

Projects

Covariate Shift Regularization using Kernel Methods
Investigated how covariate shift affects model generalization and predictive reliability under distribution mismatch; evaluated importance-weighting and self-attention-based approaches for adapting models to changing input distributions. Student project supervised by Prof. Dr. Nicole Mücke, graded 1.0.

Uncertainty Quantification in Neural Networks
Investigated uncertainty estimation in neural networks through the connection between finite-width networks and infinite-width Gaussian-process limits; compared NNGPs, deep ensembles, MC dropout and RBF Gaussian processes under distribution shift.

Binary AMR Prediction from PBP4 using Transformers
Benchmarked encoder-only pretrained transformer models, including DNABERT and Nucleotide Transformer, for binary antimicrobial resistance prediction using pbp4 genomic sequences; investigated representation learning and parameter-efficient adaptation of pretrained biological language models using PyTorch. Seminar Deep Learning in Molecular Biology, supervised by Prof. Dr. Alice C. McHardy.

Skills

Mathematical ML

Programming & Scientific Computing

Deep Learning

LLM Systems

Tools

Languages

English
Proficient

Bengali
Native

Hindi
Proficient

German
Beginner

Awards

Deutschlandstipendium Jan 2025
Awarded for 2025/26.
Deutschlandstipendium

Wipro Ltd. industry recognition, Customer Centricity Award Jan 2023
Customer Centricity Award (Q1, FY23).
Wipro Ltd.

Wipro Ltd. industry recognition, Excellence Award Jan 2019
Excellence Award (Q3–Q4, FY19).
Wipro Ltd.